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March 9, 2021

MATH 651

Midterm

1. Let X be a topological space. Let A be an arbitrary subset of X, and let

$$K = \bigcap_{\substack{C \supseteq A \\ C \text{ compact}}} C,$$

be the subset of X given by the intersection of all compact sets $C \in \{C_\alpha\}_{\alpha \in J}$ containing A.

Definition 1 (universal property of minimal compactification of A in X)**.** *We say that K satisfies the universal property of minimal compactification of A in X if for every compact $L \supseteq A$ it follows that $L \supseteq K$.*

Definition 2 (minimal compactification of A in X)**.** *If X is a topological space, and $A \subseteq X$, define the* minimal compactification of A in X *as the smallest compact subset of X which contains A, provided such a thing exists. That is, if there exists a compact $K \supseteq A$ satisfying the universal property of minimimal compactification of A in X, then we call K the minimal compactification of A in X.*

(a) Let X, A, K be as above.

Proposition 1. *i. The set K satisfies $A \subseteq C \subseteq K \subseteq X$ for any compact $C \supseteq A$.*
ii. If A has a minimal compactification in X, it must be given by K.

Proof. Consider $A \subseteq X$.

i. It is clear that K exists and satisfies $C \subseteq K \subseteq X$ for any compact $C \supseteq A$.
Since K is the intersection of a collection of sets C which each contain A as a subset, it is clear that K too contains A as a subset.

ii. Since we have already shown that K exists and satisfies $A \subseteq K \subseteq X$, we show that it satisfies the universal property of minimal compactification of A in X by considering any compact $L \supseteq A$.
Since L is a term in the intersection

$$K = \bigcap_{\substack{C \supseteq A \\ C \text{ compact} \\ C=L \text{ or } C \neq L}} C = \left[\bigcap_{\substack{C \supseteq A \\ C \text{ compact} \\ C=L}} C \right] \cap \left[\bigcap_{\substack{C \supseteq A \\ C \text{ compact} \\ C \neq L}} C \right] = L \cap \left[\bigcap_{\substack{C \supseteq A \\ C \text{ compact} \\ C \neq L}} C \right],$$

it follows that $K \subseteq L$.
If A has a minimal compactification in X, it must therefore be given by K.

In particular, we have shown that K exists and satisfies the universal property.
It may fail to be the minimal compactification of A in X, however, if it fails to be compact. □

In parts (c) and (d) we explore two categories of spaces for which compactness of K holds for some sets subsets of X but not others. First, we discuss in part (b) a category of spaces for which compactness of K is guaranteed for all subsets of X.

(b) Let $\overline{A}$ denote the closure Closure(A) of A.

Proposition 2. *i. If F is a finite subset of X, then F is compact and the minimal compactification of F in X is F itself, regardless of whether X is compact or Hausdorff.*

ii. If X is compact and Hausdorff, then for any A the minimal compactification of A in X exists and is equal to the closure $\overline{A}$.

Proof. Consider $A \subseteq K \subseteq X$.

i. Suppose A is a finite set given by $A = \{a_i\}_{i=1}^n$, and consider an open subcover $\mathcal{U} = \{U_\alpha\}_{\alpha \in J}$ in X which covers A. Then, there is an open cover $V = \{V_\alpha\}_{\alpha \in J}$ of A with the subspace topology, where $V_\alpha = U_\alpha \cap A$. Since V covers A, there are sets V_{α_i} containing a_i for each $a_i \in A$. Since V contains a finite subcover $\{V_{\alpha_i}\}_{i=1}^n$, it follows that A is compact.
Since A is the smallest compact set containing itself, it follows that $K \subseteq A$. Therefore, $A = K$ is compact. By 1.a.ii, it satisfies the universal property.
Therefore, the minimal compactification of a finite set in X is itself.

ii. Suppose X is compact and Hausdorff.
By Theorem 26.3 (p.165), any compact set C in the Hausdorff space X is closed. If A has a minimal compactification in X, then part (a) asserts that it must be given by

$$K = \bigcap_{\substack{C \supseteq A \\ C \text{ compact}}} C = \bigcap_{\substack{C \supseteq A \\ C \text{ closed}}} C = \overline{A},$$

the closure of A. Indeed, these sets exists.
If X is compact and Hausdorff, then the minimal compactification of any A in X therefore exists and is equal to $K = \overline{A}$.

□

(c) Recall that a Hausdorff space X is necessarily T_1, by Theorem 17.8 (p.99).

Proposition 3. *If X is Hausdorff but not compact, then A may or may not have a minimal compactification in X.*

Proof. Let $x \in X$, and suppose X is Hausdorff but not compact. Then, it is T_1.
We have shown in 1.b.i. that each finite set, such as $A_x = \{x\}$, has a minimal compactification of A_x in X.
We now provide an example of an open subset $A = X \setminus A_x = X \setminus \{x\}$ of X which does not have a minimal compactification.
By the definition, the minimal compactification of A must contain A, so it must be A or $A \cup \{x\} = X$. Since X is Hausdorff and A is open, it fails to be compact. Since X is not compact by hypothesis, it follows that A does not have a minimal compactification in X. □

(d)

Proposition 4. *If X is compact but not Hausdorff, then A may or may not have a minimal compactification in X.*

Proof. Let $x \in X$, and suppose X is compact but not Hausdorff.
We have shown in 1.b.i. that each finite set, such as $A_x = \{x\}$, has a minimal compactification of A_x in X.

We now provide an example of a subset A which does not have a minimal compactification in X.

If X is not Hausdorff, then the intersection K need not be compact.

Consider the line with two origins, and the non-compact subset $A = [-1, 1] - \{0_1, 0_2\}$. Its minimal compactification, if it exists, would be given by the intersection $([-1, 1] - \{0_1\}) \cap ([-1, 1] - \{0_2\})$ of compact sets. This intersection, however, turns out to be the non-compact A.

Thus, A may not have a minimal compactification in X. □

2. Let X be a topological space.

Definition 3 (limit point compactness)**.** *A space X is* limit point compact *if every infinite subset has a limit point.*

Proposition 5. *If X is T_1 and limit point compact, then every* countable *open cover of X has a finite subcover.*

Proof. Let X be T_1 and limit point compact, and consider a countable open cover $\mathcal{U} = \{U_i\}_{i=1}^{\infty}$.

Suppose $\mathcal{U}$ does not have a finite subcover.

Then, no finite union of elements of $\mathcal{U}$ can cover X. That is, each finite union

$$V_n = \bigcup_{i=1}^{n} U_n$$

has a point $x_n \notin V_n$ in its complement. Consider the subset x defined as the union

$$x = \bigcup_{n \in \mathbb{Z}_+} \{x_n\}$$

.

For each $k \in \mathbb{Z}_+$, there are infinitely many $x_i \notin U_k$, namely those for which $i \geq k$, and thus finitely many $x_i \in U_k$, namely those for which $i < n$.

By limit point compactness, there is a limit point x_0 of x.

Since $\mathcal{U}$ covers X, there is some neighborhood $U_n \in \mathcal{U}$ of x_0. Since X is T_1 and x_0 is a limit point of x, Theorem 17.9 (p.99) asserts that U_n contains infinitely many points of x.

Since we have arrived at a contradiction, it must therefore be that each open cover $\mathcal{U}$ admits a finite subcover. □

3. Let $\langle G, e, \cdot, ^{-1} \rangle$ be a topological group with a continuous maps $\text{inv} : G \to G$ and $m : G \times G \to G$ given by $\text{inv} : g \mapsto g^{-1}$ and $m : (g, h) \mapsto g \cdot h$. Let $\mathcal{U}_g$ be denote the collection of all neighborhoods of $g \in G$.

Lemma 6. *Let $gU = gU = \{gu \mid u \in U\}$, $U^{-1} = \{u^{-1} \mid u \in U\}$, and $VV = \{ab \mid a, b \in V\}$.*

(a) The collection $\mathcal{U}_g = \{gU \mid U \in \mathcal{U}_e\}$.

(b) The set $U \in \mathcal{U}_g$ if, and only if, $U^{-1} \in \mathcal{U}_{g^{-1}}$, and thus $U \in \mathcal{U}_e$ if, and only if, $U^{-1} \in \mathcal{U}_e$.

(c) If $U \in \mathcal{U}_e$, then there exists a $V \in \mathcal{U}_e$ with $VV \subseteq U$.

(a) Consider the the subset $H \subseteq G$ defined as the intersection

$$H = \bigcap_{U \in \mathcal{U}_e} U$$

of all neighborhoods of $e \in G$.

Proposition 7. *The subset H is a subgroup of G even if G is not assumed to be T_1. If G is T_1, then this subset is $H = \{e\}$.*

Proof. Clearly, $e \in H \subseteq G$. Let $a, b \in H$.
We show $ab, a^{-1} \in H$ by showing that $ab, a \in U$ for any $U \in \mathcal{U}_e$.
By Lemma 6c, there exists a $V \in \mathcal{U}_e$ with $VV \subseteq U$ and thus $Ve = eV = V \subseteq U$.
Since $V \in \mathcal{U}_e$, it follows that $H \subseteq V$ and thus that $a, b \in H \subseteq V$. Since $HH \subseteq VV \subseteq U$, it follows that $ab \in HH \subseteq U$.
By Lemma 6b, $V^{-1} \in \mathcal{U}_e$. Since $H^{-1} \subseteq V^{-1}$, it follows that $a^{-1} \in H^{-1} \subseteq V^{-1}$.
If G is T_1, then each set of the form $G - \{g\}$ is compact; and their intersection is merely $\{g\}$. □

(b) Let $AU = \{au \mid a \in A, u \in U\}$ for any subsets $A, U \subseteq G$.

Proposition 8. *For any set $A \subseteq G$, $\overline{A} = \bigcap_{U \in \mathcal{U}_e} AU$.*

Proof. Let $U \in \mathcal{U}_e$, and let $g \in \overline{A}$.
By Lemma 6b, $U^{-1} \in \mathcal{U}_e$. By Lemma 6a, $gU^{-1} \in \mathcal{U}_g$.
Since g is in the closure of A and gU^{-1} is a neighborhood of g, there is an element a in the intersection $gu^{-1} = a \in A \cap gU^{-1}$ for some $u^{-1} \in U^{-1}$.
Then, $g = gu^{-1}u = au \in AU \subseteq \overline{AU}$ and thus $g \in \bigcap_{U \in \mathcal{U}_e} AU$.
On the other hand, let $x \in \bigcap_{U \in \mathcal{U}_e} \overline{AU}$ and let $U \in \mathcal{U}_e$. Then, there exists a $V \in \mathcal{U}_e$ such that VVU. Then, $x \in \overline{AV} \subseteq AU$.
Let W be a neighborhood of x. Let $W^{-1}x \in \mathcal{U}_e$, so $x = au \in AU$ for some $a \in A$ and $u \in U$. Then, $a = xu^{-1} \in xU^{-1} = xx^{-1}W = W$. Thus, $a \in A \cap V$. Thus, every neighborhood of x contains some distinct $a \in A$. Thus, $x \in \overline{A}$. □

(c) Let $\langle G, e\rangle$ and $\langle H, e_H\rangle$ be topological groups, and let $\phi : G \to H$ be a homomorphism.

Theorem 9. *Then ϕ is continuous if, and only if, it is continuous at the identity, $e \in G$.*

Proof. Let G and H be topological groups; let $\phi : G \to H$ be a group homomorphism.
We apply Theorem 18.1 (p.104) to prove the result by showing that ϕ is continuous at the identity if, and only if, it is continuous at the point g for each $g \in G$.
If it is continuous at the point g for each $g \in G$, then this holds in particular for $e \in G$.
Suppose instead that ϕ is continuous at the identity, and consider an arbitrary neighborhood V of $\phi(g)$. By Lemma 6a, $V = \phi(g)W$ for some neighborhood W of the identity $\phi(e) = e_H \in H$. By continuity at the identity, there is a neighborhood T of e such that $\phi(T) \subseteq W$. By Lemma 6a, there is a left coset neighborhood $U = gT$ of g thus satisfying

$$\phi(U) = \phi(gT) = \phi(g)\phi(T) \subseteq \phi(g)W = V$$

due to the homomorphism property and the established equalities and set containments. Thus, ϕ is continuous at g. □

4. Let $f : X \to X$ be a function from a topological space X to itself.

Definition 4 (§23 Connected Spaces, p.148.)**.** *A space X is connected if and only if the subsets of X that are both open and closed in X are the empty set $\emptyset$ and X itself.*

Theorem 10. *If X is compact, Hausdorff, and connected, and $f : X \to X$ is continuous and open, then f is surjective.*

Proof. Suppose X is compact, Hausdorff, connected, and $f : X \to X$ is continuous and open.

If X is the empty set, then the theorem follows from the fact that $f : \emptyset \to \emptyset$ is vacuously surjective. Now suppose that X is not empty, and recall that $\emptyset \subsetneq f(X) \subseteq X$.

By Definition 4, the only non-empty subset of X that is both open and closed in X is X itself.

We show $f(X) = X$ by showing that the non-empty set $f(X)$ is both open and closed in X.

Since f is an open map and X is open in X, it follows that $f(X)$ is open in X.

Since f is continuous and X is compact, it follows that $f(X)$ is a compact subspace of X; since X is Hausdorff, it follows that $f(X)$ is closed in X.

Since $f(X)$ is open and closed, it follows that $f(X) = X$.

Therefore, f is surjective. □